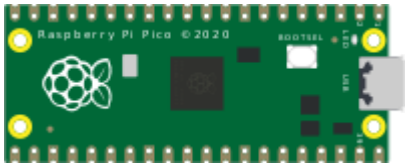
PYTHON PROGRAMMING WITH IOT

6. a)

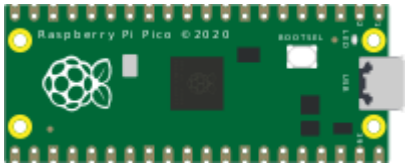


→ **Raspberry Pi Pico :** It is a small, low-cost microcontroller board used to control electronics and build DIY projects.

→ **Wokwi LED :** It is a virtual LED used in the Wokwi simulator to show how an LED would turn on and off in a circuit.

```
Connections:
  1. "pico:GP15" --- "led:A"
  2. "pico:GND.1" --- "led:C"
```

b)



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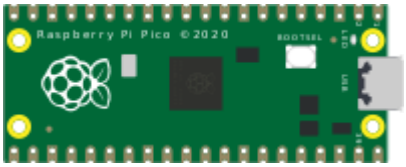
```
Connections:
  1. "pico:GP6"  --- "led:A"
  2. "pico:GND.3" --- "led:C"
```

7.



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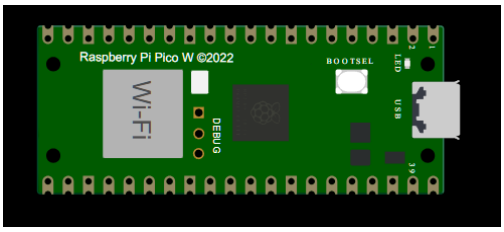
→ **Digital Humidity and Temperature Sensor :** It measures the air's moisture and temperature and sends the values as digital data.

```
Connections:
  1.  "pico:GP16" --- "dht1:SDA"
  2.  "pico:3V3" --- "dht1:VCC"
  3.  "pico:GND.1" --- "dht1:GND"
  4.  "pico:GP18" --- "redLed:A"
  5.  "pico:GND.2" --- "redLed:C"
  6.  "pico:GP20" --- "greenLed:A"
  7.  "pico:GND.3" --- "greenLed:C"
```

8.

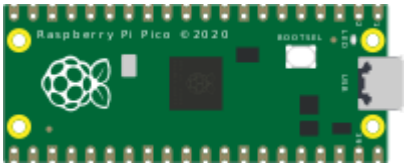


→ **Raspberry Pi Pico W :** It is a microcontroller board like the Pico, but with built-in Wi-Fi for wireless projects.

→ **Potentiometer :** It is a variable resistor you turn to change the amount of voltage in a circuit.

```
Connections:
   1. "pot1:VCC" --- "pico:3V3"
   2. "pot1:GND" --- "pico:GND.1"
   3. "pot1:SIG" --- "pico:GP26"
```

9.



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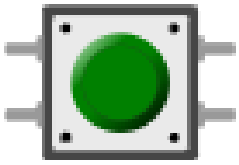
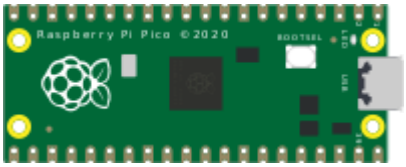


→ **Pushbutton :** It is a virtual button used in the Wokwi simulator to simulate pressing a real button in a circuit.

```
Connections:
   1. "btn1:1.r" --- "pico:GP15"
   2. "btn1:2.r" --- "pico:GND.1"
```

10.



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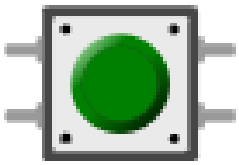

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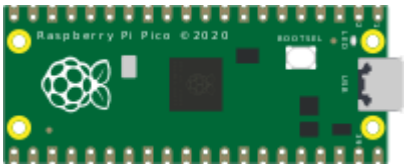
```
Connections:
1. "led1:A" --- "pico:GP0"
2. "pico:GND.1" --- "led1:C"
3. "pico:GND.4" --- "btn1:1.r"
4. "pico:GP9" --- "btn1:2.r"
5. "pico:GP9" --- "pico:GP14"
```

11.



→ **Raspberry Pi Pico :** It is a small, low-cost microcontroller board used to control electronics and build DIY projects.

→ **Digital Humidity and Temperature Sensor :** It measures the air's moisture and temperature and sends the values as digital data.

```
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1. "pico:GP16" --- "dht:SDA"
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```
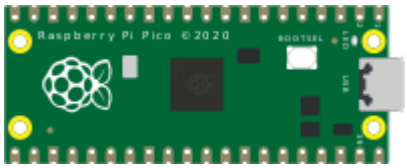

12.



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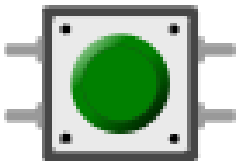
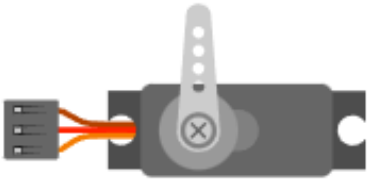


→ **Pushbutton :** It is a virtual button used in the Wokwi simulator to simulate pressing a real button in a circuit.

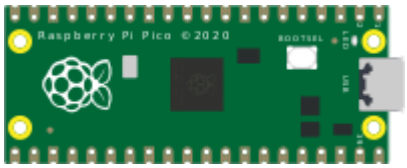
→ **Relay Module :** It is a device that lets a low-power signal safely switch a high-power circuit on or off.



→ **Servo :** It is a virtual motor in the Wokwi simulator that rotates to specific angles based on control signals.

```
Connections:
1. "pico:GP18" --- "btn1:2.l"
2. "btn1:1.r" --- "servo1:PWM"
3. "relay1:GND" --- "pico:GND.8"
4. "pico:GND.8" --- "servo1:GND"
5. "servo1:V+" --- "relay1:VCC"
```

13.



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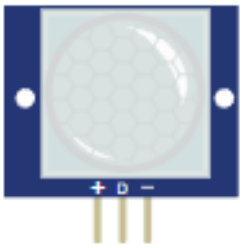
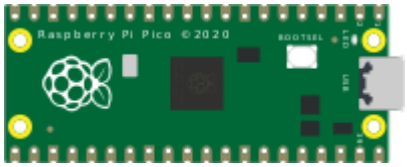


→ **Passive Infrared (PIR) motion sensor** : It detects movement by sensing changes in infrared radiation (heat) from objects, especially people.

```
Connections:
1. "pico:GND.1" --- "pir1:GND"
2. "pico:GP2" --- "pir1:OUT"
3. "pico:3V3" --- "pir1:VCC"
4. "led1:C" --- "pico:GND.3"
5. "led1:A" --- "pico:GP9"
```

14.



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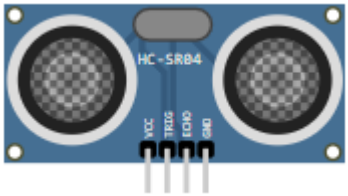
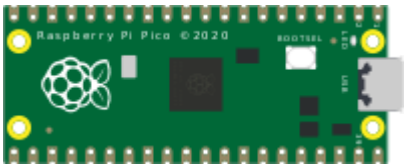


→ **HC-SR04 Ultrasonic Distance Sensor :** It measures how far away objects are by sending out sound waves and timing how long they take to bounce back.

```
Connections:
1. "ultrasonic1:GND" --- "pico:GND.1"
2. "ultrasonic1:ECHO" --- "pico:GP26"
3. "ultrasonic1:TRIG" --- "pico:GP28"
4. "ultrasonic1:VCC" --- "pico:GP5"
```

15.



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→**Seven Segment Display (TM1637) :** It is a 4-wire TM1637 module is a 7-segment LED display controlled using four wires to show numbers.

```
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2. "pico:3V3" --- "dht1:VCC"
3. "pico:GND" --- "dht1:GND"
4. "pico:GP2" --- "display:CLK"
5. "pico:GP3" --- "display:DIO"
6. "pico:3V3" --- "display:VCC"
7. "pico:GND" --- "display:GND"
```